Supplementary Materials for Manuscript Entitled

Carcinogenesis of Male Oral Submucous Fibrosis Alters Salivary Microbiomes

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Running Head: Oral Microbiome in Malignant Transformation of OSF

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Supplementary Materials and Methods

DNA recovery and qPCR. Sample groups were randomized and processed together to eliminate batch effects. Bacterial genomic DNA was isolated from 300–400 µL of saliva samples with QIAamp DNA Mini Kit (Qiagen, Germany) in accordance with the manufacturer's protocol (spin procedure). The quantity and quality of isolated gDNA were determined using a NanoDrop 2000 (Thermo Scientific, USA). The total bacteria concentrations in saliva samples were determined using SYBR-Green-based qPCR with a universal bacterial 16S rRNA primer set (27F and 518R). PCR and qPCR were performed in triplicate using a SYBR Premix Ex Taq (Takara Bio, Japan) kit on a CFX96 real-time system (Bio-Rad, USA) in accordance with previously described protocols [1].

16S rRNA gene amplicon sequencing. The V3–V4 hypervariable region of the 16S rRNA gene was amplified using a *Bacteria*-specific primer set with a v3 chemistry kit (2 × 300 bp) on a MiSeq platform (Illumina, USA). The detailed sequencing protocol has been described elsewhere [1]. In total, 4,749,181 denoizing reads were retained from 52 samples, yielding 91,330 ± 17,389 reads per library. After rarefaction to an even depth (70,675 reads), the rarefaction curves plateaued, suggesting that the obtained reads included most of the ASV types in the samples.

Bioinformatics analysis. HTS data were analyzed on the QIIME 2 platform (v2019.4) [2]. Reads were denoized using DADA2 [3], with the q2-dada2 plugin, to produce ASVs. In brief, forward and reverse primers were removed from both ends. Forward and reverse reads were truncated at 290 nt and 240 nt, respectively. After primer-sequence trimming, ASVs with lengths greater than 450 nt or less than 400 nt accounted for 1,902 reads and were excluded from the subsequent analysis. The maximum number of expected errors was two for both ends. The consensus method was used for chimera removal. Defaults were used for the other DADA2 parameters. The denoized ASVs were inserted into a reference phylogeny (Greengenes 13.8 at 99%) to develop a phylogenetic tree by using SATé-enabled phylogenetic placement (SEPP). Taxonomy was assigned to ASVs with a Scikit-learn naïve Bayesian classifier trained on the expanded Human Oral Microbiome Database (version 15.1) [4] by using the q2-feature-classifier plugin with default confidence = 0.7. Alpha diversity (observed ASVs, Pielou's evenness, Shannon's index, and Faith's PD), beta diversity (Bray–Curtis and weighted UniFrac dissimilarity matrices), and PCoA were calculated using the q2-diversity plugin at a sampling depth of 70,675 reads per sample. The stochasticity ratio was estimated using the strength of stochasticity based on different dissimilarity metrics, as described by Ning et al. [5]. The metagenome was developed in silico by using the PICRUSt2 functional prediction from the 16S rRNA genes [6] based on the MetaCyc pathway database [7]. In total, 85.77% of denoized reads were of moderate-to-high quality (nearest sequenced taxon indicator [NSTI] score < 0.15). To ensure a suitable-quality prediction, denoized reads with NSTI scores > 2 (0.37% of total denoized reads) were excluded from the PICRUSt2 analysis.

Absolute microbial load. The absolute microbial load of species *i*, A_i, was calculated as follows:

$$A_i = a_i \times C_{qPCR} \div \frac{\sum_i a_i \overline{n_i}}{\sum_i a_i}$$

where a_i is the 16S rRNA gene-copy number (GCN) normalized relative abundance for species *i* (%), $\overline{n_i}$ is the average GCN for species *i* (copies/cell), and C_{qPCR} is the qPCR-quantified bacterial concentration (copies/mL of saliva). The final term of this equation is the arithmetic mean of a sample's GCN. The GCN-normalized table was obtained using a customized QIIME 2 plugin (https://github.com/Jiung-Wen/q2-gcn-norm) with a manually curated, taxonomy-to-copy-number map file specifically for Human Oral Microbiome Database taxonomy. The 16S GCN database used in the map file was retrieved from the Ribosomal RNA Operon Copy Number Database (v5.5) [8].

Statistical analysis. The Kruskal–Wallis test was used to detect the significance of alpha diversity differences between the OSF and OSCC–OSF groups. A similarity analysis was performed to test dissimilarity in microbial composition (beta diversity) with 999 permutations. LEfSe [9] was used to identify differentially abundant species between the OSF and OSCC–OSF groups with a significance level of 0.05 for the Kruskal–Wallis test and a threshold of 2 for logarithmic LDA scores. To identify significantly different species based on absolute microbial load between the two groups, species with a prevalence greater than 33% in at least one group were selected for significance testing. We performed the Kruskal–Wallis test with a significance level of 0.05 and a threshold for the moderate effect size of 0.06. The AUC was calculated and compared using the Kruskal–Wallis test to evaluate the absolute microbial load efficiency as a diagnostic signature.

Machine learning analysis. The data input to the machine learning pipeline consisted of a feature table from the QIIME 2 workflow, alpha diversity indices, and sample metadata. The feature table consisted of 510 taxa at the species level; alpha diversity indices consisted of observed ASVs, evenness, Shannon's index, and Faith's PD. Sample metadata consisted of patient age and the use of alcohol, tobacco, and betel nut. Categorical features in sample metadata were one-hot encoded. Features with importance greater than 0.008 were selected. Scikit-learn was used to implement a random-forest algorithm and perform 5-fold cross-validation [10].

Network analysis. Correlation networks were developed based on microbial profiles at the species level by using SparCC with the QIIME 2 plugin q2-SCNIC with default settings [11]. Each species represented as a network node, and the SparCC correlation coefficient was defined as the edge weight. Signature species and their highly correlated neighbors (SparCC correlation coefficient > 0.6) were used to construct the networks. Samples with fewer than 500 read counts and species with fewer than an average of 2 read counts per sample were excluded from the analysis. Networks were visualized using Cytoscape (v3.6.1) with node color indicating phylum and edge width indicating strength of

correlation [12].

Grading of periodontal condition. Due to limited mouth opening of OSF, the periodontal condition was evaluated with panoramic radiograph by single evaluator. Digitalized panoramic films with good resolution of cementum-enamel junction and alveolar bone crest were assessed. First and second mandibular molars, or central and lateral maxillary incisors with most severe alveolar bone loss was classified according to the following categories: 0–2 mm (absence of alveolar bone loss), 2–4 mm (mild bone loss), and 4-6 mm (moderate bone loss) and ≥6mm (severe bone loss).

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Appendix TABLE 1 A list of samples, sequence reads, and diversity indices of the study participants. The samples were obtained with approval from the Institutional Review Boards of National Cheng Kung University Hospital (IRB No.: NCKUH-10304010/A-ER-102-324) and Chi Mei Medical Center (IRB No.: 10612-L02).

#	Sample ID	Center	Dataset	Oral status	Tumor location	Overall stage	Periodontal condition	Age	Non-chimeric reads	Subset Reads	Pielou_e	ASVs	Shannon	Faith's PD	16S rRNA gene (copies/mL of saliva)
1	A12	NCKUH	exploratory	OSF	--	--	mild	32	84619	70675	0.86	232	6.8	19.3	8.79E+06
2	A15	NCKUH	exploratory	OSF	--	--	severe	56	70755	70675	0.86	246	6.8	20.7	9.17E+06
3	A16	NCKUH	exploratory	OSF	--	--	NA	51	80508	70675	0.77	204	5.9	17.2	1.29E+07
4	A6	NCKUH	exploratory	OSF	--	--	mild	46	85625	70675	0.86	241	6.8	19.1	9.10E+07
5	A8	NCKUH	exploratory	OSF	--	--	edentulous	74	85871	70675	0.79	93	5.2	10.3	3.85E+07
6	B10	NCKUH	exploratory	OSF	--	--	moderate	36	72001	70675	0.80	191	6.1	19.1	7.33E+07
7	C11	NCKUH	exploratory	OSF	--	--	severe	57	77321	70675	0.80	85	5.1	9.1	2.54E+08
8	C8	NCKUH	exploratory	OSF	--	--	severe	56	129238	70675	0.78	160	5.7	14.9	3.80E+07
9	C9	NCKUH	exploratory	OSF	--	--	moderate	48	81964	70675	0.75	177	5.6	17.0	2.37E+07
10	D10	NCKUH	exploratory	OSF	--	--	mild	42	77987	70675	0.73	152	5.3	17.1	5.89E+05
11	D11	NCKUH	exploratory	OSF	--	--	mild	30	82828	70675	0.70	178	5.2	17.3	2.82E+07
12	D3	NCKUH	exploratory	OSF	--	--	NA	45	78209	70675	0.76	193	5.7	17.6	2.23E+07
13	D4	NCKUH	exploratory	OSF	--	--	NA	45	90079	70675	0.84	223	6.5	17.3	7.14E+06
14	E11	NCKUH	exploratory	OSF	--	--	moderate	52	77174	70675	0.79	165	5.9	14.7	9.54E+07
15	E13	NCKUH	exploratory	OSF	--	--	severe	41	97651	70675	0.79	188	6.0	16.7	1.03E+09
16	E14	NCKUH	exploratory	OSF	--	--	moderate	29	95938	70675	0.65	158	4.7	14.7	4.75E+07
17	E2	NCKUH	exploratory	OSF	--	--	severe	61	110115	70675	0.82	265	6.6	23.9	1.65E+07
18	F3	NCKUH	exploratory	OSF	--	--	mild	38	73826	70675	0.70	178	5.2	18.6	1.91E+07
19	A20	NCKUH	exploratory	OSCC-OSF	Buccal mucosa	I	moderate	67	87099	70675	0.79	171	5.9	14.1	8.41E+06
20	A3	NCKUH	exploratory	OSCC-OSF	Tongue	II	moderate	57	90700	70675	0.64	127	4.5	16.2	3.71E+06
21	A4	NCKUH	exploratory	OSCC-OSF	Gingiva	IV	moderate	73	78107	70675	0.82	161	6.0	18.5	7.24E+06
22	A5	NCKUH	exploratory	OSCC-OSF	Buccal mucosa	I	edentulous	64	89703	70675	0.70	108	4.7	13.0	1.35E+08
23	A7	NCKUH	exploratory	OSCC-OSF	Buccal mucosa	I	severe	59	115579	70675	0.79	201	6.0	16.1	4.74E+07
24	A1	NCKUH	exploratory	OSCC-OSF	Gingiva	IV	mild	35	80210	70675	0.78	140	5.6	13.0	8.54E+06
25	A10	NCKUH	exploratory	OSCC-OSF	Tongue	I	moderate	76	82866	70675	0.82	250	6.5	21.0	5.33E+07
26	B1	NCKUH	exploratory	OSCC-OSF	Buccal mucosa	II	moderate	56	92727	70675	0.78	130	5.5	11.9	1.88E+08
27	B11	NCKUH	exploratory	OSCC-OSF	Buccal mucosa	I	moderate	63	70791	70675	0.72	178	5.4	17.3	1.73E+08
28	B12	NCKUH	exploratory	OSCC-OSF	Buccal mucosa	I	mild	65	90830	70675	0.80	122	5.5	11.6	2.00E+07
29	B17	NCKUH	exploratory	OSCC-OSF	Gingiva	IV	severe	47	79521	70675	0.83	184	6.2	14.4	1.04E+07
30	B2	NCKUH	exploratory	OSCC-OSF	Buccal mucosa	IV	moderate	50	95907	70675	0.82	154	6.0	12.8	4.99E+08
31	B6	NCKUH	exploratory	OSCC-OSF	Soft palate	I	moderate	47	100547	70675	0.78	119	5.4	15.7	4.89E+06
32	C1	NCKUH	exploratory	OSCC-OSF	Tongue	III	mild	39	120437	70675	0.79	80	5.0	10.5	1.14E+08
33	C10	NCKUH	exploratory	OSCC-OSF	Buccal mucosa	I	moderate	60	71771	70675	0.87	147	6.3	14.2	3.88E+08
34	C13	NCKUH	exploratory	OSCC-OSF	Buccal mucosa	I	edentulous	62	84109	70675	0.62	65	3.8	9.2	4.67E+07
35	C15	NCKUH	exploratory	OSCC-OSF	Gingiva	IV	severe	51	90356	70675	0.76	263	6.1	26.0	6.09E+05
36	C17	NCKUH	exploratory	OSCC-OSF	Buccal mucosa	I	moderate	67	76327	70675	0.78	188	5.9	15.8	7.72E+07
37	C18	NCKUH	exploratory	OSCC-OSF	Buccal mucosa	I	mild	53	110665	70675	0.82	243	6.5	20.6	2.57E+07
38	C2	NCKUH	exploratory	OSCC-OSF	Buccal mucosa	I	mild	58	136375	70675	0.76	205	5.8	17.2	3.15E+07
39	C4	NCKUH	exploratory	OSCC-OSF	Buccal mucosa	I	severe	39	77921	70675	0.84	211	6.5	18.4	3.43E+08
40	C6	NCKUH	exploratory	OSCC-OSF	Gingiva	III	moderate	75	78923	70675	0.76	101	5.1	11.2	5.27E+07
41	C7	NCKUH	exploratory	OSCC-OSF	Tongue	I	moderate	46	110466	70675	0.74	188	5.6	17.4	1.51E+08
42	D17	NCKUH	exploratory	OSCC-OSF	Gingiva	II	mild	57	83930	70675	0.86	173	6.4	14.9	1.04E+07

43	D19	NCKUH	exploratory	OSCC-OSF	Buccal mucosa	I	edentulous	65	81071	70675	0.81	78	5.1	9.1	6.48E+07
44	D5	NCKUH	exploratory	OSCC-OSF	Tongue	II	severe	64	101798	70675	0.71	132	5.0	14.0	3.78E+08
45	D6	NCKUH	exploratory	OSCC-OSF	Palate	I	moderate	55	88147	70675	0.82	125	5.7	11.9	9.21E+07
46	D7	NCKUH	exploratory	OSCC-OSF	Gingiva	IV	edentulous	77	130776	70675	0.76	108	5.1	11.8	7.74E+06
47	D8	NCKUH	exploratory	OSCC-OSF	Buccal mucosa	III	mild	50	129114	70675	0.76	151	5.5	16.1	1.01E+06
48	D9	NCKUH	exploratory	OSCC-OSF	Tongue	II	mild	34	124309	70675	0.76	182	5.7	14.4	7.46E+06
49	E18	NCKUH	exploratory	OSCC-OSF	Gingiva	IV	moderate	69	73800	70675	0.77	78	4.9	10.3	4.22E+07
50	E4	NCKUH	exploratory	OSCC-OSF	Buccal mucosa	I	moderate	68	77582	70675	0.85	211	6.6	18.9	7.62E+06
51	E6	NCKUH	exploratory	OSCC-OSF	Soft palate	II	moderate	70	88809	70675	0.67	118	4.6	13.4	9.16E+07
52	E7	NCKUH	exploratory	OSCC-OSF	Buccal mucosa	III	moderate	62	106199	70675	0.58	49	3.2	7.5	1.36E+08
53	OSF-100	CMMC	validation	OSF	--	--	NA	64	121582	70675	0.88	300	7.2	24.4	5.23E+09
54	OSF-1042	CMMC	validation	OSF	--	--	NA	27	96537	70675	0.76	168	5.6	16.0	4.05E+09
55	OSF-1062	CMMC	validation	OSF	--	--	NA	25	123077	70675	0.70	156	5.1	12.7	4.02E+08
56	OSF-217	CMMC	validation	OSF	--	--	NA	21	105239	70675	0.79	164	5.8	14.4	9.95E+09
57	OSF-262	CMMC	validation	OSF	--	--	NA	49	131850	70675	0.86	297	7.1	23.5	1.45E+09
58	OSF-285	CMMC	validation	OSF	--	--	NA	34	83656	70675	0.74	170	5.5	17.8	2.11E+07
59	OSF-365	CMMC	validation	OSF	--	--	NA	21	106183	70675	0.85	203	6.5	16.7	1.57E+09
60	OSF-570	CMMC	validation	OSF	--	--	NA	51	111689	70675	0.85	254	6.8	21.7	1.17E+09
61	OSF-741	CMMC	validation	OSF	--	--	NA	31	150455	70675	0.86	269	6.9	21.5	7.42E+09
62	OSF-797	CMMC	validation	OSF	--	--	NA	28	111831	70675	0.79	188	6.0	16.7	1.02E+08
63	SCC-1012	CMMC	validation	OSCC-OSF	NA	I	NA	74	160693	70675	0.73	157	5.4	16.7	1.64E+09
64	SCC-1035	CMMC	validation	OSCC-OSF	NA	II	NA	44	126683	70675	0.82	202	6.3	17.5	5.65E+08
65	SCC-519	CMMC	validation	OSCC-OSF	NA	IV	NA	38	104463	70675	0.70	177	5.2	15.9	6.29E+08
66	SCC-579	CMMC	validation	OSCC-OSF	NA	II	NA	56	127026	70675	0.85	247	6.7	23.3	3.88E+08
67	SCC-673	CMMC	validation	OSCC-OSF	NA	I	NA	58	162932	70675	0.71	132	5.0	15.0	2.03E+09
68	SCC-718	CMMC	validation	OSCC-OSF	NA	I	NA	49	164276	70675	0.75	226	5.8	19.9	8.27E+09
69	SCC-887	CMMC	validation	OSCC-OSF	NA	II	NA	66	115851	70675	0.82	240	6.5	21.6	8.04E+09
70	SCC-937	CMMC	validation	OSCC-OSF	NA	II	NA	43	103825	70675	0.68	205	5.2	19.4	2.62E+08
71	SCC-938	CMMC	validation	OSCC-OSF	NA	I	NA	66	134427	70675	0.50	120	3.5	12.9	9.64E+09
72	SCC-982	CMMC	validation	OSCC-OSF	NA	II	NA	51	74325	70675	0.82	204	6.3	19.2	9.38E+08
73	SCC-990	CMMC	validation	OSCC-OSF	NA	I	NA	54	121949	70675	0.82	244	6.5	26.3	1.47E+09

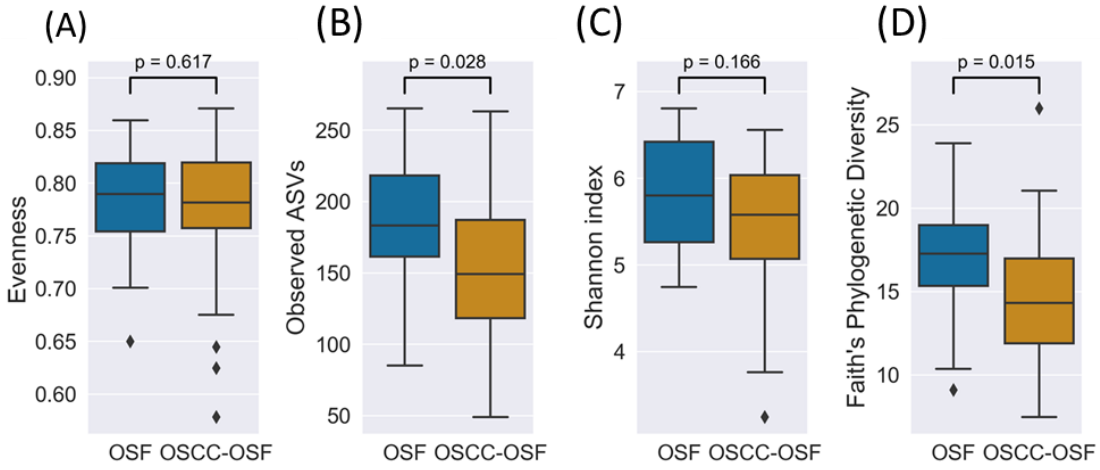
Noted: NA, not available.

Appendix TABLE 2 Relative abundance in terms of average and standard deviation (Stdev) and prevalence of signature species candidates analyzed using LEfSe.

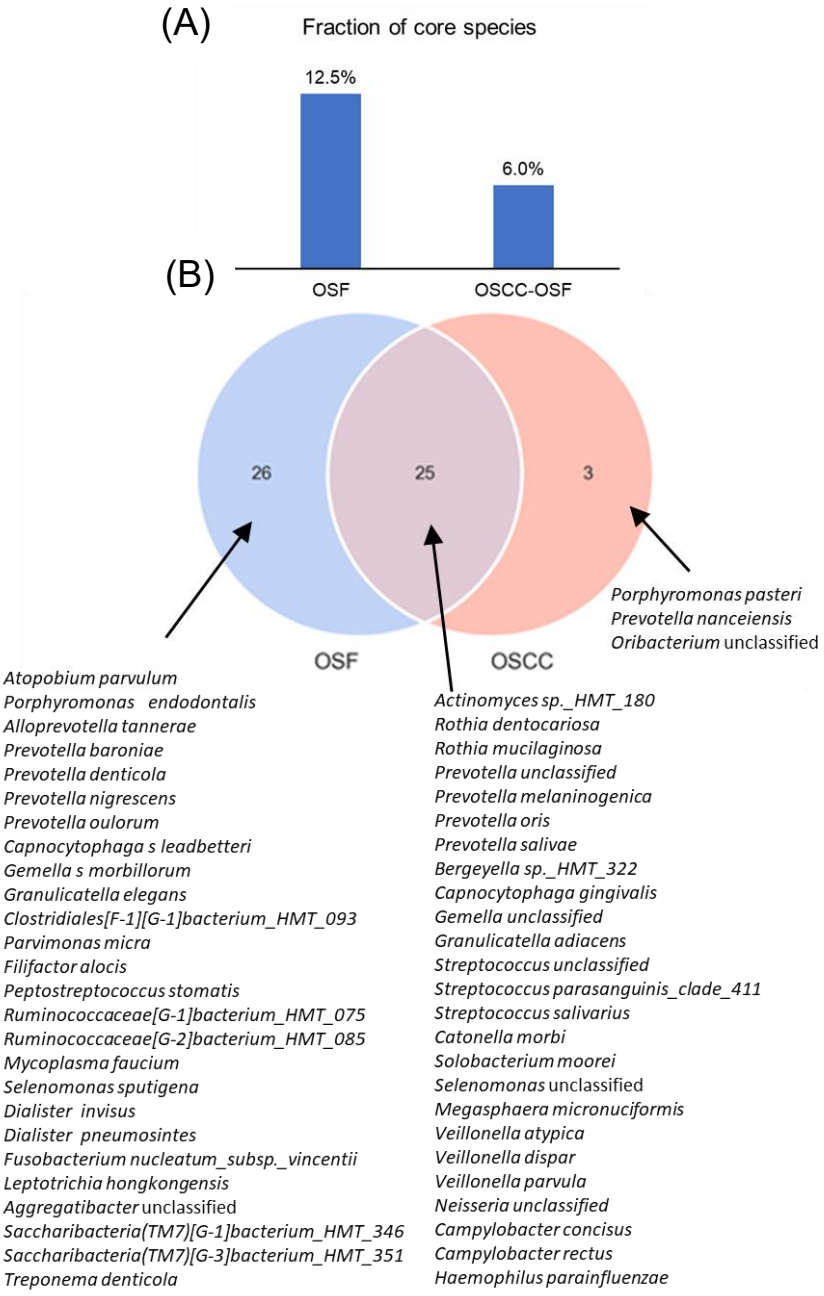
Species taxa	OSF				OSCC-OSF			
	Average (%)	Stdev (%)	Prevalence (%)	Quartile	Average (%)	Stdev (%)	Prevalence (%)	Quartile
Haemophilus pittmaniae	0.33	1.28	11.1	Q1	1.03	2.87	47.1	Q2
Prevotella sp. HMT-309	0.0009	0.039	5.6	Q1	0.15	0.4	32.4	Q2
Treponema sp. HMT-270	0.03	0.05	50	Q3	0.032	0.109	17.6	Q1
Treponema sp. HMT927	0.67	1.45	55.6	Q3	0.02	0.08	8.8	Q1
Mycoplasma faucium	0.67	1.46	77.8	Q4	0.28	0.89	26.5	Q2
Prevotella denticola	0.55	0.93	77.9	Q4	0.13	0.34	41.2	Q2
Mollicutes sp. HMT-504	0.297	0.011	27.8	Q2	0	0	0	Q1
Mitsuokella sp. HMT-131	0.2	0.42	44.4	Q2	0.01	0.03	14.7	Q1
Absconditabacteria (SR1) sp. HMT-874	0.417	0.694	61.1	Q3	0.33	1.02	26.5	Q2
Clostridiales sp. HMT-093	0.314	0.603	83.3	Q4	0.208	0.542	38.2	Q2
Prevotella baroniae	0.241	0.373	83.3	Q4	0.1	0.203	41.2	Q2
Treponema sp. HMT-237	0.28	0.377	55.6	Q3	0.183	0.499	23.5	Q1
Prevotella oulorum	0.441	0.005	94.4	Q4	0.303	0.005	58.8	Q3
Prevotella multisaccharivorax	0.173	0.314	50	Q3	0.012	0.03	17.6	Q1
Selenomonas sputigena	0.375	0.003	100	Q4	0.25	0.004	64.7	Q3
Eubacterium saphenus	0.178	0.003	61	Q3	0.045	0.001	23.5	Q2
Trepotema sp. HMT-238	0.142	0.351	33	Q2	0.03	0.14	5.9	Q1
Actinomyces sp. HMT-169	0.269	0.432	61.1	Q3	0.088	0.2	26.5	Q2
Selenomonas dianae	0.113	0.002	27.8	Q2	0.0062	0.00026	5.9	Q1
Catonella sp. HMT-451	0.152	0.254	55.6	Q3	0.066	0.221	17.6	Q1
Capylobacter sp. HMT-044	0.109	0.003	27.8	Q2	0.0146	0.0007	5.9	Q1
Porphyromonas catoniae	0.113	0.152	50	Q3	0.026	0.103	11.8	Q1
Treptonema sp. HMT-258	0.109	0.274	44.4	Q2	0.023	0.065	14.7	Q1
Selenomonas HMT-501	0.008	0.0002	11.1	Q1	0	0	0	Q1
Dialister micraerophilus	0.059	0.001	22.2	Q1	0.0007	4.10E-05	2.9	Q1
Prevotella sp. HMT-300	0.105	0.129	61.1	Q3	0.028	0.065	35.3	Q2
Treponema sp. HMT-235	0.003	0.0001	11.1	Q1	0	0	0	Q1
Bacteroidetes sp. HMT-505	0.035	0.001	11.1	Q1	0	0	0	Q1
Lachnospiraceae sp. HMT-163	0.008	0.0002	27.8	Q2	0.0002	1.20E-05	2.9	Q1
Parascardovia denticolens	0.004	0.0001	11.1	Q1	0	0	0	Q1
Prevotella sp. HMT-314	0.03	0.001	16.7	Q1	0	0	0	Q1
Mollicutes sp. HMT-906	0.053	0.102	44.4	Q2	0.011	0.03	11.8	Q1
Bifidobacteriaceae sp. HMT-407	0.003	0.0001	11.1	Q1	0	0	0	Q1
Prevotella sp. HMT-315	0.032	0.066	44.4	Q2	0.002	0.011	2.9	Q1
Megasphaera sp. HMT-123	0.028	0.001	16.7	Q1	0	0	0	Q1
Prevotella HMT-942	0.028	0.001	22.2	Q1	0.0005	2.70E-05	2.9	Q1
Peptostreptococcaceae sp. HMT-369	0.047	0.082	66.7	Q3	0.02	0.07	26.5	Q2
Bacteroidetes sp. HMT-280	0.028	0.071	38.9	Q2	0.008	0.026	11.8	Q1
Prevotella enoeca	0.012	0.0001	27.8	Q2	0.0025	0.0001	5.9	Q1
Fretibacterium HMT-362	0.036	0.001	27.8	Q2	0.02	0.0008	5.9	Q1
Treponema sp. HMT-260	0.014	0.0001	16.7	Q1	0	0	0	Q1
Peptostretococcaceae sp. HMT-091	0.016	0.0001	27.8	Q2	0.001	4.70E-05	8.8	Q1

Note: Quartile, Q4 (> 75%); Q3(50~75%); Q2(25~50%);Q1(< 25%).

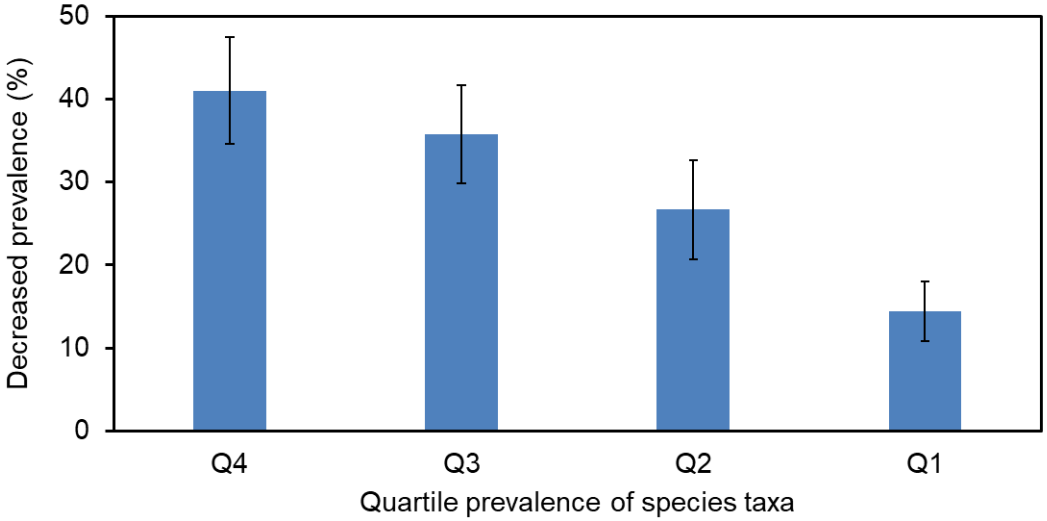
The taxa in bold blue are signature species detected with the microbial load approach.



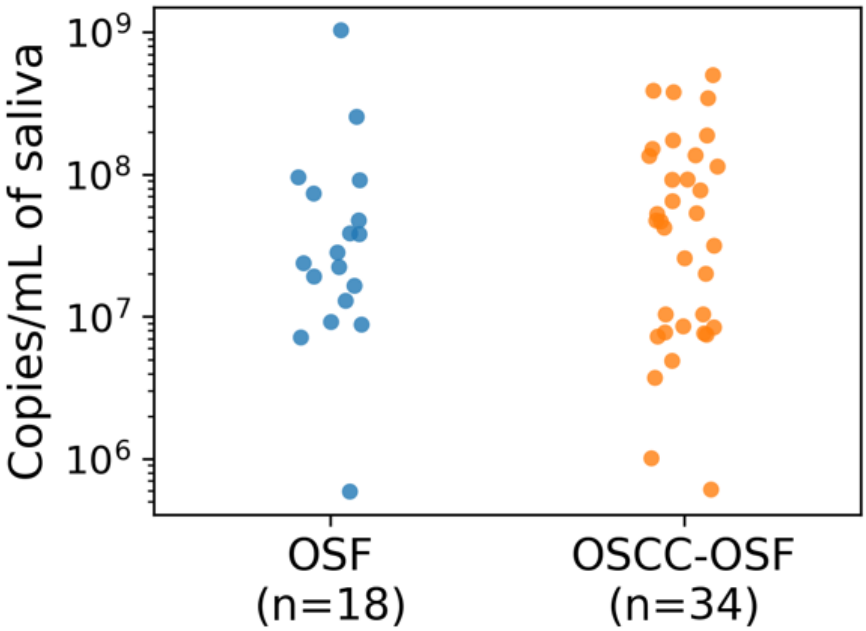
Appendix FIG 1. Boxplots of differences in amplicon sequence variant (ASV) diversity between the oral submucous fibrosis (OSF) (n = 18) and oral squamous cell carcinoma (OSCC)–OSF (n = 34) groups. The Kruskal–Wallis test was used to determine whether the differences in average values obtained from the two groups were statistically significant. The *p*-value is displayed at the top of each panel. (A) Pielou's evenness index. (B) Observed ASVs. (C) Shannon index. (D) Faith's phylogenetic diversity.



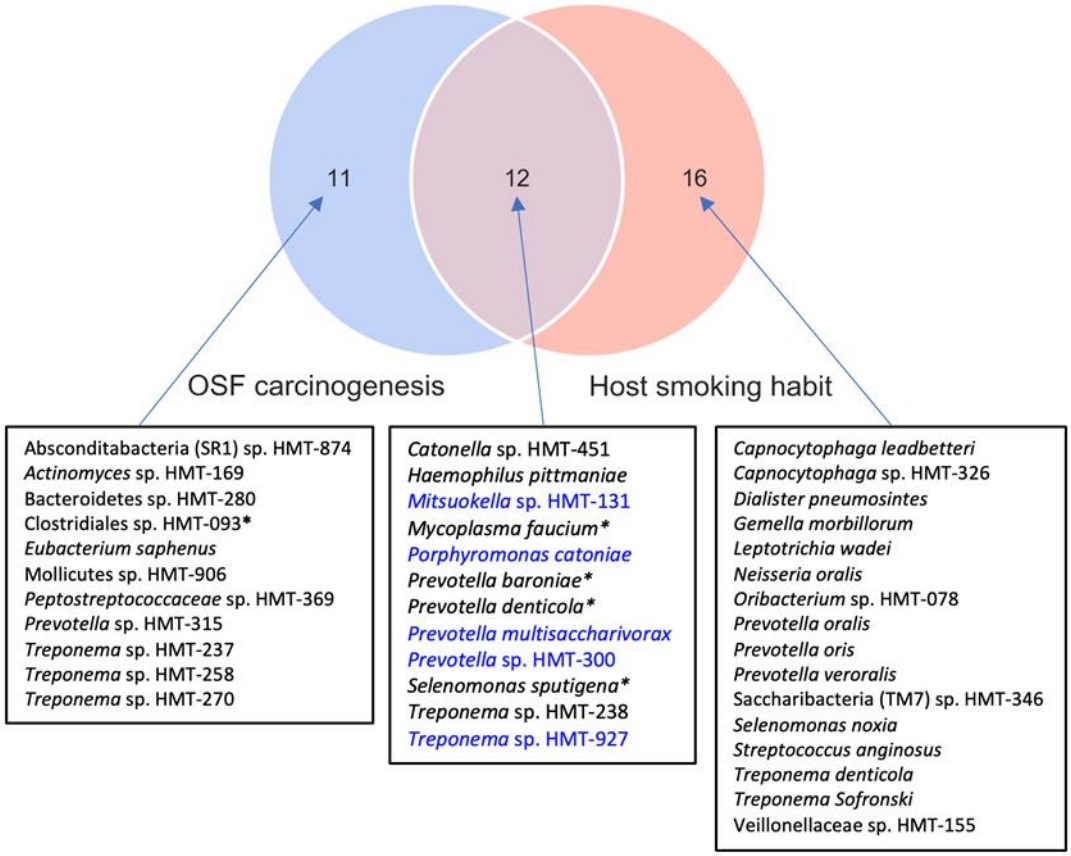
Appendix FIG 2. (A) Fractions of core species with a frequency >75% in the OSF and OSCC-OSF sample groups. (B) Venn diagram analysis of core species detected in the OSF and OSCC-OSF cohorts, respectively. The names of the core species were listed below the Venn diagram.



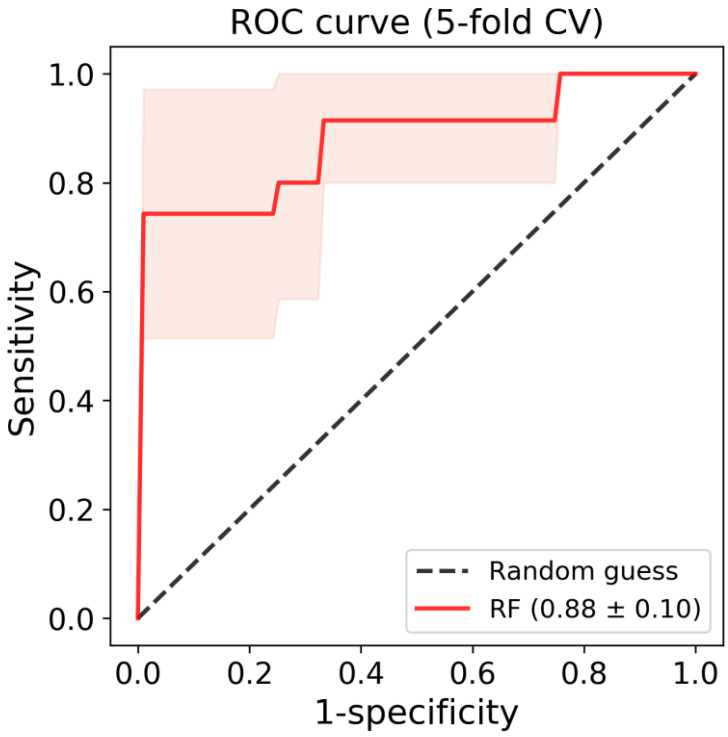
Appendix FIG 3. Distribution of taxa prevalence reduced in the association of OSF malignancy in different prevalence ranks of OSF samples. Q4, >75% of sample prevalence, Q3, 50~75% of sample prevalence, Q2, 25~50% of sample prevalence, and Q1, <25% of sample prevalence.



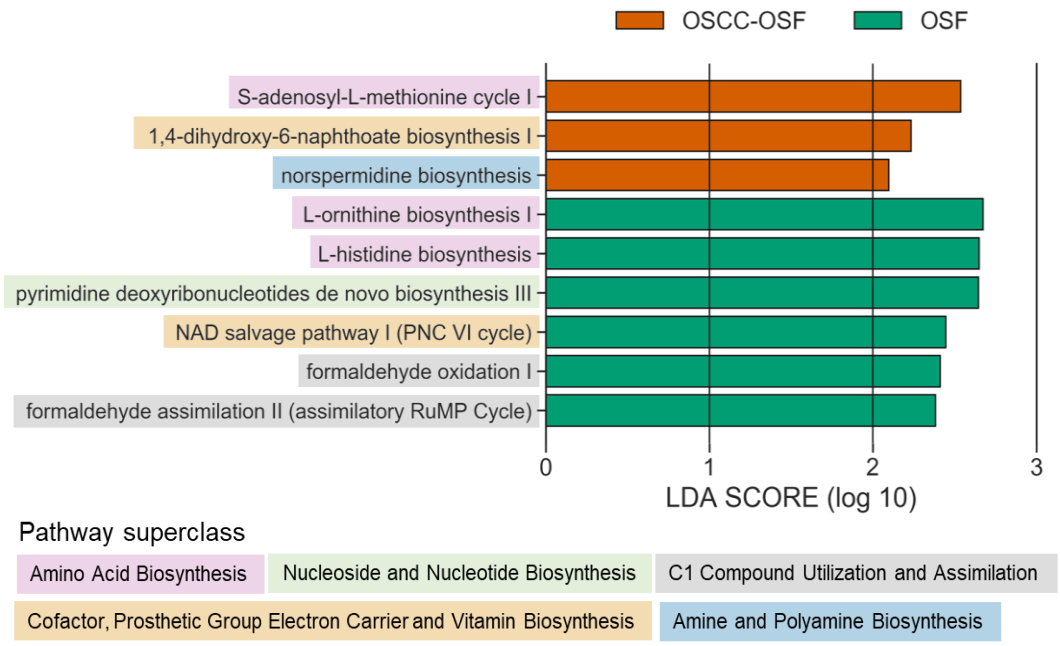
Appendix FIG 4. Data distribution of using the quantitative polymerase chain reaction method to analyse the copies of total bacterial 16S rRNA gene in the saliva of OSF and OSCC-OSF cohorts.



Appendix FIG 5. Venn diagram analysis of species taxa associated with OSF carcinogenesis and smoking habits. The species taxa were detected in accordance with absolute microbial loads that were significantly different in the two groups (Kruskal-Wallis test, $p<0.05$). The names of the species were listed below the Venn diagram. The taxa labeled with an asterisk (*) are core species of the OSF cohort. The taxa highlighted in blue were signature species detected by multiple methods (see the main text). The abundance of OSF carcinogenesis-related taxa between patients with mild periodontal disease (n = 13) and moderate-to-severe disease (n = 31) did not differ significantly (Kruskal–Wallis test, $P>0.105$). Edentulous patients (n = 5) were excluded from this analysis. The periodontal status of three patients was not available.



Appendix FIG 6. The ROC curve of random forest algorithm with an input of selected species taxa, Faith's PD, and current smoking habits under the 5-fold cross validation of exploratory dataset.



Appendix FIG 7. Linear discriminant analysis (LDA) effect size ($\alpha = 0.05$, $\log_{10}$LDA score > 2.0) identified differentially abundant metabolic pathways as predicted using PICRUSt2, between the oral submucous fibrosis (OSF) and oral squamous cell carcinoma (OSCC)–OSF groups. The pathway face color indicates its superclass.